THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	Carries out of the columns: [1, 0, 1, 0].
2	**(c)** 6	Divisors of 44: 1, 2, 4, 11, 22, 44.
3	**(b)** 2	Pairs: $(90, 82) \to (82, 8) \to (74, 8) \to (66, 8) \to (58, 8) \to (50, 8) \to (42, 8) \to (34, 8) \to (26, 8) \to (18, 8) \to (10, 8) \to (8, 2) \to (6, 2) \to (4, 2) \to (2, 2) \to (2, 0)$.
4	**(a)** 2	$146 = 3 \times 48 + 2$.
5	**(a)** $1, 2, \ldots, n$	Every j from 1 to n is tested.
6	**(b)** 6	Pairs: $(90, 50) \to (50, 40) \to (40, 10) \to (30, 10) \to (20, 10) \to (10, 10) \to (10, 0)$.
7	**(a)** 449	Apply the rule each time.
8	**(d)** 3	Pairs: $(138, 117) \to (117, 21) \to (96, 21) \to (75, 21) \to (54, 21) \to (33, 21) \to (21, 12) \to (12, 9) \to (9, 3) \to (6, 3) \to (3, 3) \to (3, 0)$.
9	**(a)** 1	Carries out of the columns: [0, 1, 0, 0].
10	**(a)** 4	Divisors of 77: 1, 7, 11, 77.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(54, 30) \to (30, 24) \to (24, 6) \to (18, 6) \to (12, 6) \to (6, 6) \to (6, 0)$ (1); 6 subtractions, gcd $= 6$ (1).	1+1
12	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(70, 42) \to (42, 28) \to (28, 14) \to (14, 14) \to (14, 0)$ (1); 4 subtractions, gcd $= 14$ (1).	1+1
13	Check each j from 1 to 42: 42 numbers are checked; List $= [1, 2, 3, 6, 7, 14, 21, 42]$ (1); It has 8 entries (1).	1+1
14	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(55, 40) \to (40, 15) \to (25, 15) \to (15, 10) \to (10, 5) \to (5, 5) \to (5, 0)$ (1); 6 subtractions, gcd $= 5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(92, 72) \to (72, 20) \to (52, 20) \to (32, 20) \to (20, 12) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$ (1); 8 subtractions, gcd $= 4$ (1).	1+1+1
16	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(195, 165) \to (165, 30) \to (135, 30) \to (105, 30) \to (75, 30) \to (45, 30) \to (30, 15) \to (15, 15) \to (15, 0)$ (1); 8 subtractions, gcd $= 15$ (1).	1+1+1
17	Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: tens, hundreds, thousands (1); $2860 + 7672 = 10532$ (1).	1+1+1
	OR: Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(112, 98) \to (98, 14) \to (84, 14) \to (70, 14) \to (56, 14) \to (42, 14) \to (28, 14) \to (14, 14) \to (14, 0)$ (1); 8 subtractions, gcd $= 14$ (1).	1+1+1
18	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(165, 135) \to (135, 30) \to (105, 30) \to (75, 30) \to (45, 30) \to (30, 15) \to (15, 15) \to (15, 0)$ (1); 7 subtractions, gcd $= 15$ (1).	1+1+1
	OR: Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: units, tens (1); $5199 + 3314 = 8513$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rule: new $x = 2x - 2$ (1); Values: 5, 8, 14, 26, 50, 98 (2); Final $x = 98$ (2).	1+2+2
	OR: Digits picked (from the right): 6, 9, 6, 5 (1); Sum $= 6 + 9 + 6 + 5$ (2); Final $s = 26$ (the digit sum) (2).	1+2+2
20	Rule: new $x = 2x - 1$ (1); Values: 9, 17, 33, 65, 129 (2); Final $x = 129$ (2).	1+2+2
	OR: Digits picked (from the right): 0, 6, 8, 4 (1); Sum $= 0 + 6 + 8 + 4$ (2); Final $s = 18$ (the digit sum) (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	Pairs: $(116, 80) \to (80, 36) \to (44, 36) \to (36, 8) \to (28, 8) \to (20, 8) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$.
2	**(d)** 3	$351 = 12 \times 29 + 3$.
3	**(a)** 48	gcd(g, 0) = g.
4	**(c)** 7	Pairs: $(114, 66) \to (66, 48) \to (48, 18) \to (30, 18) \to (18, 12) \to (12, 6) \to (6, 6) \to (6, 0)$.
5	**(d)** 450	Apply the rule each time.
6	**(d)** 2	Pairs: $(170, 138) \to (138, 32) \to (106, 32) \to (74, 32) \to (42, 32) \to (32, 10) \to (22, 10) \to (12, 10) \to (10, 2) \to (8, 2) \to (6, 2) \to (4, 2) \to (2, 2) \to (2, 0)$.
7	**(b)** 2	Carries out of the columns: [1, 1, 0, 0].
8	**(b)** 2	Divisors of 89: 1, 89.
9	**(b)** 16	Pairs: $(112, 96) \to (96, 16) \to (80, 16) \to (64, 16) \to (48, 16) \to (32, 16) \to (16, 16) \to (16, 0)$.
10	**(a)** 1	$208 = 3 \times 69 + 1$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(70, 60) \to (60, 10) \to (50, 10) \to (40, 10) \to (30, 10) \to (20, 10) \to (10, 10) \to (10, 0)$ (1); 7 subtractions, gcd $= 10$ (1).	1+1
12	Check each j from 1 to 49: 49 numbers are checked; List $= [1, 7, 49]$ (1); It has 3 entries (1).	1+1
13	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(44, 32) \to (32, 12) \to (20, 12) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$ (1); 6 subtractions, gcd $= 4$ (1).	1+1
14	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(112, 48) \to (64, 48) \to (48, 16) \to (32, 16) \to (16, 16) \to (16, 0)$ (1); 5 subtractions, gcd $= 16$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(88, 64) \to (64, 24) \to (40, 24) \to (24, 16) \to (16, 8) \to (8, 8) \to (8, 0)$ (1); 6 subtractions, gcd $= 8$ (1).	1+1+1
16	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(180, 135) \to (135, 45) \to (90, 45) \to (45, 45) \to (45, 0)$ (1); 4 subtractions, gcd $= 45$ (1).	1+1+1
17	Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: hundreds, thousands (1); $9541 + 4847 = 14388$ (1).	1+1+1
	OR: Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(140, 90) \to (90, 50) \to (50, 40) \to (40, 10) \to (30, 10) \to (20, 10) \to (10, 10) \to (10, 0)$ (1); 7 subtractions, gcd $= 10$ (1).	1+1+1
18	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(160, 96) \to (96, 64) \to (64, 32) \to (32, 32) \to (32, 0)$ (1); 4 subtractions, gcd $= 32$ (1).	1+1+1
	OR: Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: units, tens (1); $4445 + 3199 = 7644$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rule: new $x = 2x - 5$ (1); Values: 8, 11, 17, 29 (2); Final $x = 29$ (2).	1+2+2
	OR: Digits picked (from the right): 2, 2, 0, 4 (1); Sum $= 2 + 2 + 0 + 4$ (2); Final $s = 8$ (the digit sum) (2).	1+2+2
20	Rule: new $x = 2x - 3$ (1); Values: 2, 1, -1, -5, -13 (2); Final $x = -13$ (2).	1+2+2

	OR: Digits picked (from the right): 2, 2, 1, 5 (1); Sum $= 2 + 2 + 1 + 5$ (2); Final $s = 10$ (the digit sum) (2).	1+2+2

Mistakes to avoid:

- Forgetting to add the carry from the previous column in column addition.
- Counting a carry when the column sum is exactly 9 (a carry needs a sum of 10 or more).
- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
- In Euclid's subtraction method, subtracting the larger number from the smaller or stopping before the two numbers are equal.
- Reading $m \bmod n$ as the quotient of $m \div n$; it is the remainder, and it is always less than n.
- Skipping steps when tracing an algorithm, so that a variable's value is not updated in the table.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the numbers n and $m - n$	Replace by the smaller number and the difference.
2	**(a)** 4	Pairs: $(88, 66) \to (66, 22) \to (44, 22) \to (22, 22) \to (22, 0)$.
3	**(c)** 18	Apply the rule each time.
4	**(d)** 8	Pairs: $(184, 168) \to (168, 16) \to (152, 16) \to (136, 16) \to (120, 16) \to (104, 16) \to (88, 16) \to (72, 16) \to (56, 16) \to (40, 16) \to (24, 16) \to (16, 8) \to (8, 8) \to (8, 0)$.
5	**(b)** 2	Carries out of the columns: [1, 0, 0, 1].
6	**(c)** 9	Divisors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36.
7	**(a)** 2	Pairs: $(114, 34) \to (80, 34) \to (46, 34) \to (34, 12) \to (22, 12) \to (12, 10) \to (10, 2) \to (8, 2) \to (6, 2) \to (4, 2) \to (2, 2) \to (2, 0)$.
8	**(b)** 5	$233 = 12 \times 19 + 5$.
9	**(c)** a finite sequence of well-defined steps	Finiteness.
10	**(a)** 6	Pairs: $(81, 63) \to (63, 18) \to (45, 18) \to (27, 18) \to (18, 9) \to (9, 9) \to (9, 0)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Check each j from 1 to 31: 31 numbers are checked; List $= [1, 31]$ (1); It has 2 entries (1).	1+1
12	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(63, 39) \to (39, 24) \to (24, 15) \to (15, 9) \to (9, 6) \to (6, 3) \to (3, 3) \to (3, 0)$ (1); 7 subtractions, gcd $= 3$ (1).	1+1
13	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(58, 40) \to (40, 18) \to (22, 18) \to (18, 4) \to (14, 4) \to (10, 4) \to (6, 4) \to (4, 2) \to (2, 2) \to (2, 0)$ (1); 9 subtractions, gcd $= 2$ (1).	1+1
14	Check each j from 1 to 52: 52 numbers are checked; List $= [1, 2, 4, 13, 26, 52]$ (1); It has 6 entries (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(116, 68) \to (68, 48) \to (48, 20) \to (28, 20) \to (20, 8) \to (12, 8) \to (8, 4) \to (4, 4) \to (4, 0)$ (1); 8 subtractions, gcd $= 4$ (1).	1+1+1
16	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(180, 100) \to (100, 80) \to (80, 20) \to (60, 20) \to (40, 20) \to (20, 20) \to (20, 0)$ (1); 6 subtractions, gcd $= 20$ (1).	1+1+1
17	Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: hundreds, thousands (1); $7950 + 2826 = 10776$ (1).	1+1+1
	OR: Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(102, 60) \to (60, 42) \to (42, 18) \to (24, 18) \to (18, 6) \to (12, 6) \to (6, 6) \to (6, 0)$ (1); 7 subtractions, gcd $= 6$ (1).	1+1+1
18	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first (1); $(189, 153) \to (153, 36) \to (117, 36) \to (81, 36) \to (45, 36) \to (36, 9) \to (27, 9) \to (18, 9) \to (9, 9) \to (9, 0)$ (1); 9 subtractions, gcd $= 9$ (1).	1+1+1
	OR: Add from the units column, carry 1 when a column sum exceeds 9 (1); Columns with a carry: tens, hundreds, thousands (1); $6953 + 5580 = 12533$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rule: new $x = 2x - 4$ (1); Values: 9, 14, 24, 44 (2); Final $x = 44$ (2).	1+2+2
	OR: Digits picked (from the right): 2, 9, 0, 5 (1); Sum $= 2 + 9 + 0 + 5$ (2); Final $s = 16$ (the digit sum) (2).	1+2+2
20	Rule: new $x = 2x - 4$ (1); Values: 8, 12, 20, 36, 68 (2); Final $x = 68$ (2).	1+2+2
	OR: Digits picked (from the right): 0, 9, 9, 2 (1); Sum $= 0 + 9 + 9 + 2$ (2); Final $s = 20$ (the digit sum) (2).	1+2+2

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- Leaving out the divisors 1 and the number itself when listing divisors, or listing a divisor twice (as in $36 = 6 \times 6$).
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